



Cairo University
 Faculty of Engineering
 Credit Hours System
 Communications and Computer Engineering Department

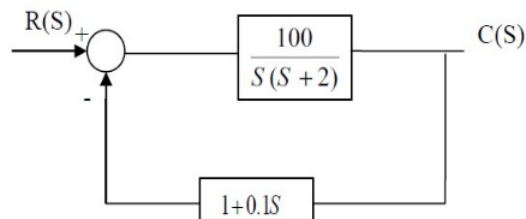
Fall 2026

Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
 Course TA Eng. Hassan El-Menier

Sheet 4: Transient Response

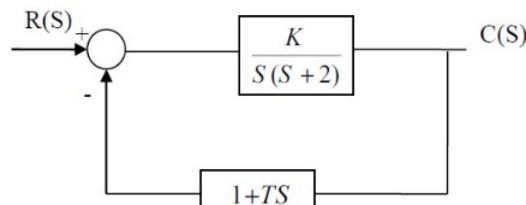
Question (1):

The block diagram of a feedback control system is shown in the following figure. Determine the values of ζ , ω_n , M_p and t_S .



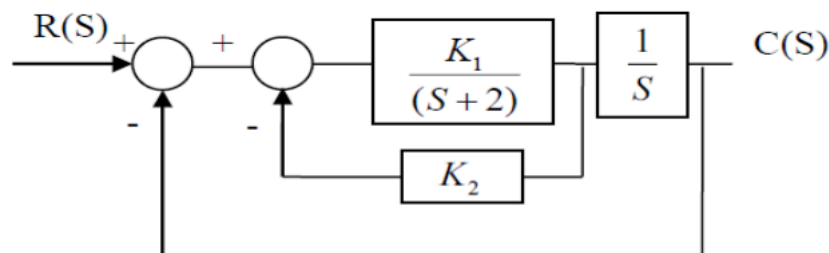
Question (2):

Referring to the system shown in figure, determine the values of K and T such that the system has a $\zeta = 0.7$, and $\omega_n = 4$ rad/sec



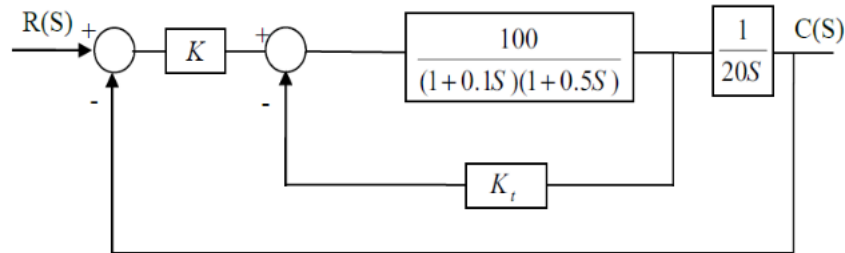
Question (3):

Referring to the system shown in figure, determine the values of K_1 and K_2 such that the system has a maximum overshoot in unit step response is 25% and the peak time is 2 sec.



Question (4):

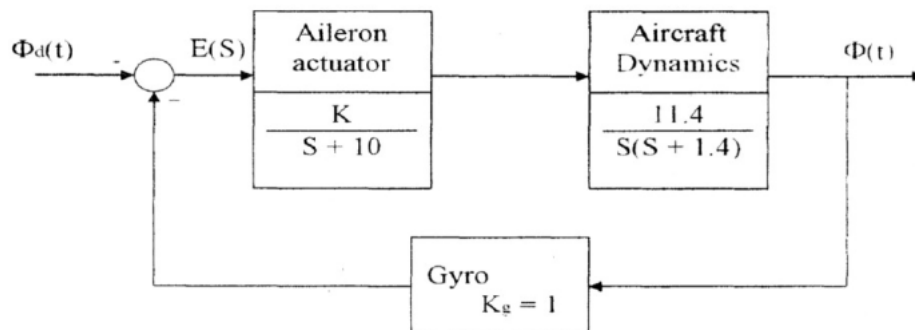
For the control system shown in the figure:



1. Determine the values of K and K_t such that the output has a maximum overshoot 4.3% and the rise time is approximately 2 sec.
2. With the values of K and K_t obtained, find the steady state error when the input is a unit ramp function.

Question (5):

The roll control autopilot of a jet fighter is shown in the below figure. The goal is to select a suitable K so that the response to a unit step command $\phi_d(t)$ will provide a response $\phi(t)$ that is a fast response and has an overshoot of less than or equal 9.5%:



1. Using the concept of dominant poles, find a suitable value of K that will achieve the desired transient response. Predict the transient response of the system (i.e. get t_r , t_p and t_s).
2. Find the static error coefficients of the system. Evaluate the value of K that gives minimum steady state error for a unit ramp input.